04

Four graded items, and today's homework

Leave a full 20 minutes for this. Students must leave today knowing exactly what they owe.

Four graded items, 100 marks in total

Assessment	Format	Weight	Due	Deliverable	What the marking mainly looks at
Early-stage assessment	Team	10%	After today's class; the class rep emails everything in at the start of Lesson 2	A 200-word English Word document with at least 3 verifiable citations	Tests one ability only posing a good problem
Mid-term assessment	Individual	20%	Weekend after Lesson 2; presented end of class over three sessions from Lesson 3	One HTML game, a complete scene tree in the system, a 100–150 word English document, a 5-page deck, and a 3-minute video	The topic must be your own dream career, and it must contain real setbacks, not only rewards
Final project	Team	50%	Ongoing; presented in Lesson 7	A complete project that runs in a browser, exports to one offline file, has at least eight selectable named objects, and uses four AI entry points	Not whether the demo looks pretty, but whether you defined what "done" means before you started, then honestly showed whether you got there
Class participation	Individual	20%	All six lessons	Your contribution on discussion cards, your hands-on link each lesson, and the questions you ask on presentation day	This is not an attendance mark. The quality of the questions you ask during other teams' final presentations counts here too

One thing to note: the final is a team deliverable only, individual work ends at the mid-term. So in the last week you can put all your energy into one thing.

Today's homework

- **In the domain your team was assigned, pose one problem that genuinely needs solving and is worth solving. The method is the four value questions.**
- Deliverable: a 200-word English Word document with at least 3 citations.
  - 200 words is short, so every sentence has to earn its place. That is the point: it forces you to narrow the problem.
  - The 3 citations are there to make you go and look things up. Literature, industry data, or real cases with a source, but not your own guesses.
  - Suggested budget: two or three sentences on who hits what difficulty in what situation, one or two on magnitude, two on why existing solutions fall short, one on why it needs 3D or AI, and a final one on what counts as solved.
- **Citations must be traceable. The team that receives your problem will rely on them to understand the background.**

- Deadline: **TODAY**, 18 Aug 2026, 23:59, late = 0 marks

Seven domains, seven teams

A · Education

Time spent on lesson prep and marking, abstract concepts that are hard to teach, experiments limited by equipment

B · Architecture

Communicating a design, construction briefings, clients who cannot read drawings

C · Industry (Robotic)

Assembly training time, remote repair, veteran know-how that is hard to pass on

D · Games

Content production cost, how long it takes to validate a mechanic, retention

E · Healthcare

Few chances to rehearse surgery, poor rehab adherence, doctor-patient communication

F · E-commerce

Return rates, sizes and results that are hard to predict, long decision chains

G · Tourism and culture

Peak and off-peak seasons, ruins that cannot be recreated, tours that all feel identical

Challenge track

Use the A100 server I provide to reproduce this paper as closely as you can: https://omnifacerig.github.io/

Seven domains, seven teams

Group Number	Given Name	Surname	Preferred Name	Gender (M/F)	Year of Study	
1	Jiayue	Guan	Guan	M	2	Class Rep
	Xinglei	Pang	Pang	M	3	
	Zhiyun	He	Zoey	F	1	
	Luojie	Yang	Roger	M	1	
2	Siyi	Li	Kath	F	2	Sub-group Leader
	Yunzhu	Chen	Yunzhu	F	3	
	Chenyu	He	Chenyu	M	1	
	Yanjia	Chen	Alice	F	1	
3	Qiwei	Zhao	Zhao	M	3	Sub-group Leader
	Yuanhao	Jin	Jin	M	2	
	Huaixi	Liu	Hwai	M	1	
	Zihan	Wang	Zihan	M	2	
4	Jingcheng	Shou	Shou	F	2	Sub-group Leader
	Xiaoheng	Li	Sean	M	1	
	Haohan	Deng	Jasper	M	3	
	Jiaming	Guo	Mina	F	1	
5	Yishan	Liu	Nina	F	1	Sub-group Leader
	Yuxi	Chen	Jason	M	2	
	Weiyi	He	Wei	M	3	
	Danyang	Qian	Qian	M	2	
6	Zhihao	Yao	Yao	M	3	Sub-group Leader
	Xiangyu	Guo	Guo	M	2	
	Quanze	Li	Li	M	1	
	Yuhui	Zheng	Hardy	M	2	
7	Qishu	Jin	Bruce	M	3	Sub-group Leader
	Rui	Su	Rafer	M	1	
	Yuhan	Zhang	Octo	M	2	

Coming up: build a game about your dream career

- **The point is to grow yourself towards that career, not to recreate a workplace scene for other people to look at.**
  - Recreate a workplace and the protagonist is the job; what I am asking for has you as the protagonist.
- Step one is not designing a game. It is writing down honestly: what I want to become, what I am missing now, and which part of that gap can be closed by doing a little every day.
- Example: if you want to start a company, first analyse where you are and what you lack, then design fun knowledge levels, check in daily to study startup material, mix in tests along the way, and pull in real case videos from the web.
- **Hard requirement: there must be real setbacks, not only rewards.**
  - If anything you do passes, then passing carries no information, and you learn nothing about whether you can actually do it.
- Playable in under three minutes. Presented at the end of class across three sessions from Lesson 3, about 10 people per session, 3 minutes each, marked live by the instructor.

- Deadline: **Sunday, 23 Aug** 2026, 23:59, late = 0 marks

Coming up: build a game about your dream career

Mid-term deliverables (five items):

- 1) A game playable in under 3 minutes, packaged as a single HTML file;
- 2) It must display a complete scene tree inside the AI multimodal system, and it must be demoable and shareable;
- 3) A 100–150 word English document (Word);
- 4) A 5-page deck;
- 5) A 3-minute video.

- Deadline: **Sunday, 23 Aug** 2026, 23:59, late = 0 marks

Build a working 3D or AI application with the system

- It must run in a browser and export to a single file that opens offline.
- **The scene needs at least eight individually selectable named objects, meaning you hand in a live scene, not a rendered image.**
- Use at least four of the system's AI entry points (text, image, video, 3D), and in your presentation say clearly what problem each one solved.
- Presentations are in final session: 9 minutes per team plus 4 minutes of questions, and every team member must come on stage and speak.
- **If part of it did not get built, say plainly "we did not manage this part, and here is why". That scores higher than hiding it.**

- Deadline: **Wednesday 26 Aug** 2026, 23:59, late = 0 marks

Build a working 3D or AI application with the system

Final deliverables (five items):

- 1) One complete project folder containing at most 5 HTML files;
- 2) It must display a complete scene tree inside the AI multimodal system, and be demoable and shareable; the project must include all of these: images, video, audio, 3D models, avatar models, avatar animation, and a world model.
- 3) An 800–1000 word English document, with the prompts you used pasted in an appendix (prompts do not count towards the word limit);
- 4) A 20-page deck;
- 5) A 3-minute video.

- Deadline: **Wednesday 26 Aug** 2026, 23:59, late = 0 marks

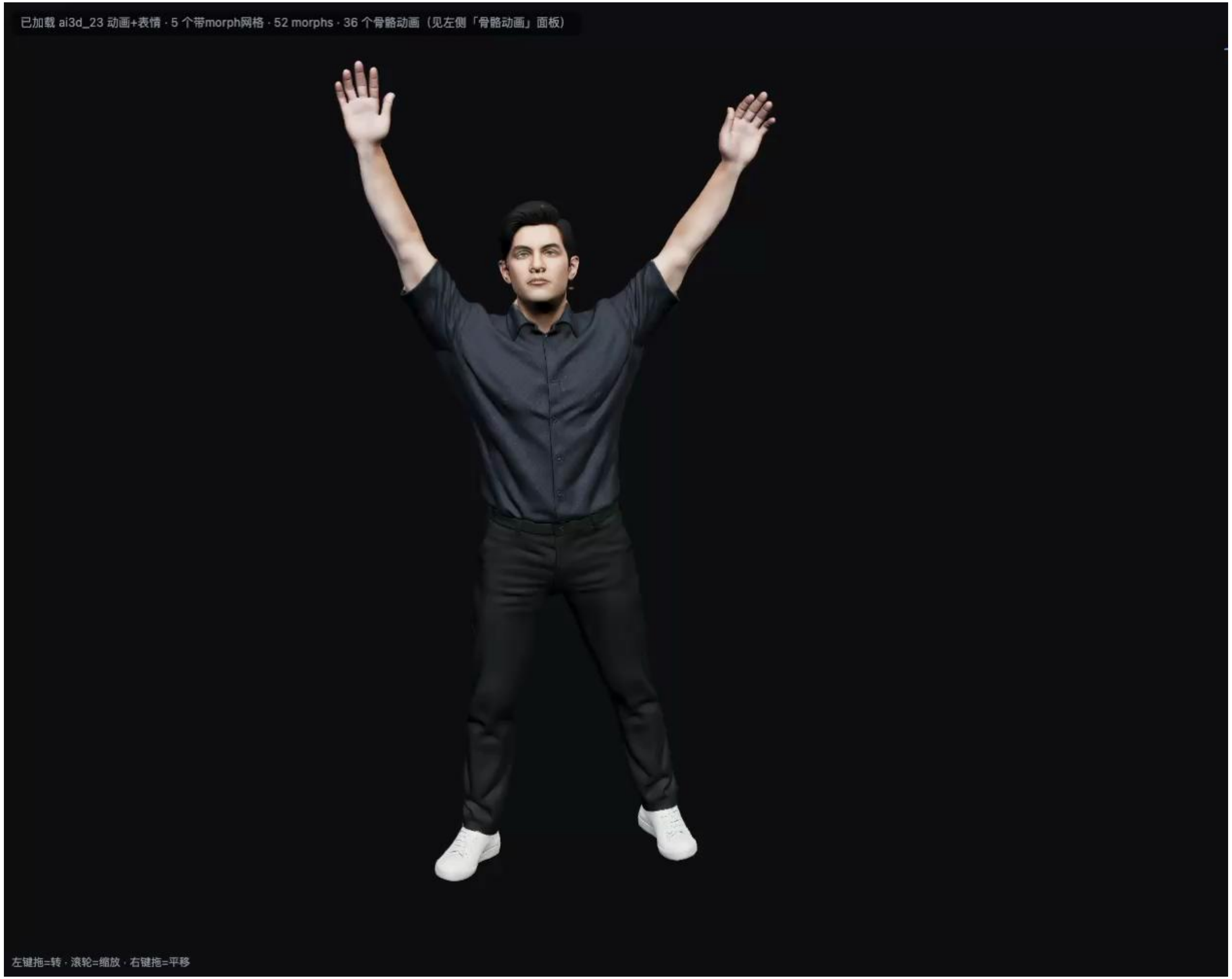
Intensive research-paper reproduction challenge (form your own team of 5–9 people)

Deliverables (five items):

- 1) A complete A100 server project, so anyone can preview and try the whole workflow from a link
- 2) Meet the requirements of https://omnifacerig.github.io/ : upload an image, then output a character with a motion skeleton + facial expression animation, as a glb file
- 3) No broken facial expressions, and the teeth and mouth interior must be positioned and animated correctly
- 4) Support the ARKit 52 blendshape set
- 5) Lip-sync for both Chinese and English audio, correctly aligned

Resources provided:

- 1) A100 server
- 2) Several high-res T-pose avatar glb models, plus 2D T-pose avatar images
- 3) A walkthrough of parts of the existing solution's code


已加载 ai3d_23 动画+表情 · 5 个带morph网格 · 52 morphs · 36 个骨骼动画（见左侧「骨骼动画」面板）
左键拖=转 · 滚轮=缩放 · 右键拖=平移